



Lesson 6.10 — Modeling with Exponentials and Logs

Big idea: When data doubles or halves on a regular schedule — or you're asked "how long until ..." — exponentials and logs are the right tool. The variable in the exponent is what logs were invented to find.

Quick review

Set up: $A(t) = A_0 \cdot b^t$ or $A_0 e^{kt}$.

Find the rate or k using a known data point.

Solving for time: isolate the exponential, take $\ln$ (or $\log$), divide.

Continuous vs. discrete: use e^{kt} for continuous (instantaneous) processes; use $(1 \pm r)^t$ for periodic (annual, monthly).

Worked example

Worked example. Bacteria triple every 4 hours. How long to grow from 100 to 10,000?

$A(t) = 100 \cdot 3^{t/4}$. Set $100 = 10,000 / 3^{t/4}$: $3^{t/4} = 100$. $t/4 = \log_3(100) \approx 4.19$. $t \approx 16.77$ hours.

Warm-ups

1. A population grows by 8% per year. Write $A(t)$ starting from 5,000.
2. A radioactive sample halves every 12 hours. Write $A(t)$ starting from 100 g.
3. An investment \$1,000 at 5% continuously compounded — write $A(t)$.

Practice

1. A radioactive isotope decays at 5%/yr. Half-life?
2. A bacteria culture grows from 200 to 1,200 in 5 hours. Find growth model and predict population at 8 hours.
3. Cooling: $T(t) = 70 + 50e^{-0.1t}$ ($^{\circ}\text{F}$, t in minutes). When does T reach 80° ?

Challenge

1. A loan of \$10,000 has interest compounded continuously at 6%. After how many years does the loan balance exceed \$20,000?
2. Newton's Law of Cooling: A coffee cup cools from 200°F to 150°F in 5 minutes in a 70°F room. Estimate temperature 15 minutes after the start.



Answer key — Lesson 6.10

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $A(t) = 5000(1.08)^t$.
2. $A(t) = 100(0.5)^{t/12}$.
3. $A(t) = 1000 e^{0.05t}$.

Practice

1. $0.5 = (0.95)^t \Rightarrow t = \ln 0.5 / \ln 0.95 \approx 13.51$ years.
2. $1200/200 = 6 = b^5 \Rightarrow b = 6^{1/5} \approx 1.4310$. $A(t) = 200(1.4310)^t$. $A(8) \approx 200(8.585) \approx 1,717$.
3. $80 = 70 + 50e^{-0.1t} \Rightarrow e^{-0.1t} = 0.2 \Rightarrow -0.1t = \ln 0.2 \Rightarrow t \approx 16.09$ min.

Challenge

1. $20000 = 10000 e^{0.06t} \Rightarrow \ln 2 = 0.06t \Rightarrow t \approx 11.55$ years.
2. Use $T(t) = 70 + 130 e^{-kt}$. At $t = 5$: $150 = 70 + 130e^{-5k} \Rightarrow e^{-5k} = 80/130 \Rightarrow k = (\ln(130/80))/5 = (\ln 1.625)/5 \approx 0.0972$. $T(15) = 70 + 130 e^{-0.0972 \cdot 15} = 70 + 130 e^{-1.458} \approx 70 + 130(0.2326) \approx 100.2$ °F.